2D Critical-Field Estimate of Trench-MOSFET Breakdown

Green-function derivation, three grouped V_{BD}–N_D sweeps and corner sensitivity

Reproducible analytical-series model; no empirical curve fitting

Interpretation. This is an exact Green-series solution of a *prescribed linear, fully depleted, local-oxide* Poisson problem, combined with numerical 2D critical-field minimization. It is **not** validated physical BV_{DSS}. Artificial corner exclusion is **not** the same as a rounded trench, and the results vary strongly with the exclusion distance.

1 Silicon domain, electrostatic PDE, and boundary conditions

The depleted silicon half-cell is the rectangle $\Omega = (0, L) \times (0, a)$, where x runs from source to drain and y runs from the symmetry plane toward the trench oxide. Fixed, uniform donor density gives

$$-(\phi_{xx} + \phi_{yy}) = k, \qquad k = \frac{qN_D}{\varepsilon_s}. \tag{1}$$

The distinct drain V_D and field-plate V_{FP} excitations enter through

$$\phi(0, y) = 0, \qquad \phi(L, y) = V_D, \tag{2}$$

$$\phi_y(x, 0) = 0, \qquad \phi_y(x, a) + \frac{\phi(x, a)}{b} = \frac{V_{\mathrm{FP}}}{b} + \frac{\sigma_0}{\varepsilon_s}, \tag{3}$$

where $b = \varepsilon_s t_{\mathrm{ox}}/\varepsilon_{\mathrm{ox}}$ is in length units and $\sigma_0 = qN_{\mathrm{ox}}$ is the uniform interfacial sheet charge. In the source/field-plate and drain/field-plate corners, ideal boundary prescriptions can be incompatible for their first derivatives.

2 Common Green function and modal coefficients

Let $P = (x, y)$, $Q = (\xi, \eta)$. Define $-\nabla_Q^2 G(P, Q) = \delta(P - Q)$ with homogeneous Dirichlet data on $\xi = 0, L$, Neumann data on $\eta = 0$ and Robin data on $\eta = a$. The Dirichlet eigenfunctions in the source–drain direction are $\sin(\alpha_n x)$, $\alpha_n = n\pi/L$. Set

$$S_n = \frac{2[1 - (-1)^n]}{n\pi}, \qquad R_n = \frac{2(-1)^n}{n\pi}, \tag{4}$$

$$\Delta_n = \alpha_n \sinh(\alpha_n a) + \frac{\cosh(\alpha_n a)}{b}, \qquad Q_n(y) = \frac{\cosh(\alpha_n y)}{b\Delta_n}. \tag{5}$$

The longitudinal Green factor in the *transverse* coordinate is

$$p_n(y) = \cosh(\alpha_n y), \tag{6}$$

$$q_n(y) = \cosh[\alpha_n(a - y)] + \frac{\sinh[\alpha_n(a - y)]}{\alpha_n b}, \tag{7}$$

$$g_n(y, \eta) = \frac{p_n(\min(y, \eta))q_n(\max(y, \eta))}{\Delta_n}. \tag{8}$$

It satisfies $g_n'(0, \eta) = 0$, $g_n'(a, \eta) + g_n(a, \eta)/b = 0$ and derivative jump $g_n'(\eta^+, \eta) - g_n'(\eta^-, \eta) = -1$. Therefore

$$G(P, Q) = \frac{2}{L} \sum_{n=1}^{\infty} \sin(\alpha_n x) \sin(\alpha_n \xi) g_n(y, \eta). \tag{9}$$

The drain voltage is a Dirichlet boundary input and enters through $-V_D \int_0^a G_\xi(P; L, \eta) \mathrm{d}\eta$; the plate voltage and interface charge enter through $G(P; \xi, a)$; the uniform space charge enters through $\int_\Omega G(P, Q) \mathrm{d}Q$.

3 Four independent potential terms and uniform-charge identity

For the uniform sheet-charge case, define its *effective Robin voltage*

$$V_{\text{eff}} = V_{\text{FP}} + \frac{b\sigma_0}{\varepsilon_s} = V_{\text{FP}} + \frac{bqN_{\text{ox}}}{\varepsilon_s}. \tag{10}$$

The exact series solution for $0 < x < L$ is

$$\boxed{\phi(x,y) = V_D\frac{x}{L} + \sum_{n=1}^{\infty}\sin(\alpha_n x)\left[\frac{kS_n}{\alpha_n^2}(1-Q_n(y)) + Q_n(y)\big(V_D R_n + V_{\text{eff}}S_n\big)\right].} \tag{11}$$

Thus the independent transfer functions are

$$F_D(x,y) = \frac{x}{L} + \sum_{n\geq 1} R_n \sin(\alpha_n x)Q_n(y), \tag{12}$$

$$F_{\text{FP}}(x,y) = \sum_{n\geq 1} S_n \sin(\alpha_n x)Q_n(y), \tag{13}$$

$$H_\rho(x,y) = \sum_{n\geq 1}\frac{S_n}{\alpha_n^2}\sin(\alpha_n x)(1-Q_n(y)), \tag{14}$$

$$H_\sigma(x,y) = bF_{\text{FP}}(x,y), \quad \text{for spatially uniform } \sigma_0. \tag{15}$$

Accordingly $\phi = V_D F_D + V_{\text{FP}}F_{\text{FP}} + kH_\rho + (\sigma_0/\varepsilon_s)H_\sigma$. At this level of the model, uniform N_{ox} and uniform V_{FP} are not separately identifiable from potential alone because they share F_{FP}.

4 Analytical 2D field-gradient response functions

The physical field is $\boldsymbol{E} = -\nabla\phi$. Since the breakdown condition uses its magnitude, define the potential gradients $\boldsymbol{d} = \nabla F_D$, $\boldsymbol{p} = \nabla F_{\text{FP}}$, $\boldsymbol{h} = \nabla H_\rho$; changing all signs does not change the magnitude. Let

$$Q_n'(y) = \frac{\alpha_n \sinh(\alpha_n y)}{b\Delta_n}. \tag{16}$$

Then differentiate the exact potential series:

$$d_x = \frac{1}{L} + \sum_{n\geq 1}\alpha_n R_n \cos(\alpha_n x)Q_n(y), \qquad d_y = \sum_{n\geq 1} R_n \sin(\alpha_n x)Q_n'(y), \tag{17}$$

$$p_x = \sum_{n\geq 1}\alpha_n S_n \cos(\alpha_n x)Q_n(y), \qquad p_y = \sum_{n\geq 1} S_n \sin(\alpha_n x)Q_n'(y), \tag{18}$$

$$h_x = \frac{L}{2} - x - \sum_{n\geq 1}\frac{S_n}{\alpha_n}\cos(\alpha_n x)Q_n(y), \qquad h_y = -\sum_{n\geq 1}\frac{S_n}{\alpha_n^2}\sin(\alpha_n x)Q_n'(y). \tag{19}$$

The identity $\sum_{n\geq 1}(S_n/\alpha_n)\cos(\alpha_n x) = L/2 - x$ is used to accelerate the donor-field evaluation away from corners. The gradient separates as

$$\nabla\phi = V_D\boldsymbol{d} + \boldsymbol{u}, \qquad \boldsymbol{u} = V_{\text{eff}}\boldsymbol{p} + k\boldsymbol{h}. \tag{20}$$

At $y = 0$, the transverse gradient vanishes and one recovers the previously used centerline criterion, but this does not locate the maximum over the entire silicon region.

5 Exact local root and two-dimensional search, with no fitting

At a fixed silicon point, the E_c criterion is

$$|V_D\boldsymbol{d} + \boldsymbol{u}|^2 = E_c^2. \tag{21}$$

Introduce $A = \boldsymbol{d}\cdot\boldsymbol{d}$, $B = \boldsymbol{d}\cdot\boldsymbol{u}$, and $C = \boldsymbol{u}\cdot\boldsymbol{u}$. One obtains

$$AV_D^2 + 2BV_D + C - E_c^2 = 0. \tag{22}$$

Provided $A > 0$ and $C < E_c^2$ at every considered point, its *first positive crossing* is

$$\boxed{V_c(x,y) = \frac{-B(x,y) + \sqrt{B(x,y)^2 + A(x,y)[E_c^2 - C(x,y)]}}{A(x,y)}.} \tag{23}$$

For a chosen evaluation region, the predicted threshold is

$$\boxed{V_{\mathrm{BD}}^{(2D,r_c)} = \min_{(x,y)\in\Omega_{r_c}} V_c(x,y).} \tag{24}$$

If $\max_{\Omega_{r_c}} |\boldsymbol{u}| \geq E_c$ at $V_D = 0$, the dataset records a missing value rather than inventing a positive breakdown margin.

Corner exclusion. The ideal rectangular model has incompatible mixed-boundary prescriptions near the silicon/oxide corners, so field maxima can be nonphysical or singular. The sampled region is

$$\Omega_{r_c} = \overline{\Omega} \setminus \left[B_{r_c}((0,a)) \cup B_{r_c}((L,a))\right] \tag{25}$$

using exclusion radii $r_c = 0.05$, 0.10, and 0.15 μm. Exclusion is a *numerical diagnostic*; it does not impose the rounded-corner electrostatics of a real trench.

6 Numerical settings and grouped results

The series is evaluated in μm (field units V/μm), with $L = 1.25\,\mu$m, $a = 0.425\,\mu$m, $\varepsilon_s = 11.7\varepsilon_0$, $\varepsilon_{\mathrm{ox}} = 3.9\varepsilon_0$, and a fixed illustrative critical field $E_c = 30\,\mathrm{V}/\mu\mathrm{m} = 0.30\,\mathrm{MV/cm}$. We sample N_D from 10^{15} to $8 \times 10^{16}\,\mathrm{cm}^{-3}$ using 80 values, 420 Fourier modes and a 111×47 position grid. All numbers are computed directly from the series, without curve fitting.

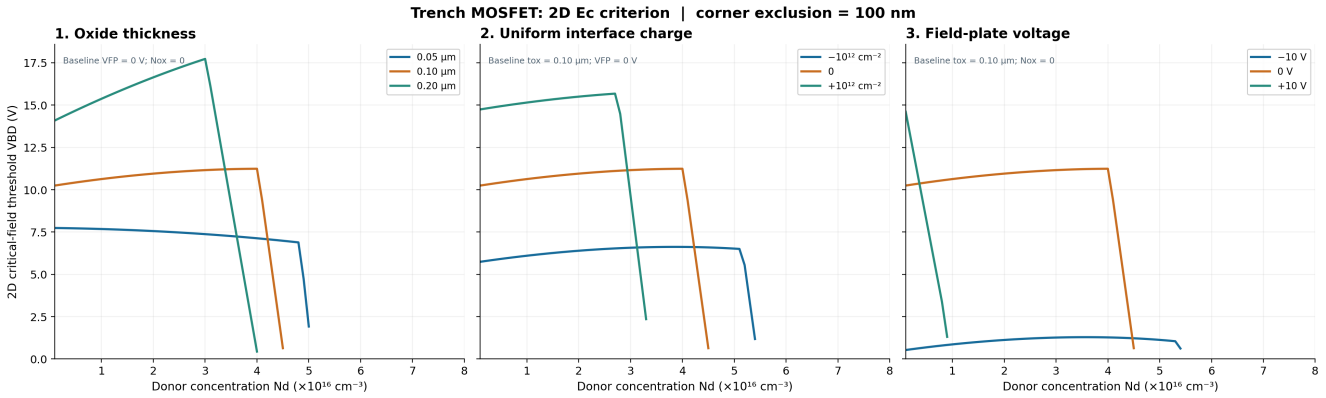


Figure 1: Baseline $r_c = 0.10\,\mu$m: separate thickness, uniform charge and field-plate voltage sweeps. Missing portions indicate E_c was reached at $V_D = 0$ in the evaluated region.

Table 1: Illustrative 2D thresholds at $N_D = 10^{16}\,\mathrm{cm}^{-3}$, $r_c = 0.10\,\mu$m.

Sweep	Input	$V_{\mathrm{BD}}^{(2D)}$ (V)
Oxide thickness	$t_{\mathrm{ox}} = 0.05\,\mu$m	7.677
	$0.10\,\mu$m	10.627
	$0.20\,\mu$m	15.369
Fixed charge	$N_{\mathrm{ox}} = -10^{12}\,\mathrm{cm}^{-2}$	6.099
	$N_{\mathrm{ox}} = 0$	10.627
	$N_{\mathrm{ox}} = +10^{12}\,\mathrm{cm}^{-2}$	15.148
Field plate	$V_{\mathrm{FP}} = -10\,\mathrm{V}$	0.861
	$V_{\mathrm{FP}} = 0$	10.627
	$V_{\mathrm{FP}} = +10\,\mathrm{V}$	already $\geq E_c$ at $V_D = 0$

7 Critical location and cutoff sensitivity

At baseline $N_D = 10^{16}\,\mathrm{cm}^{-3}$, $t_{ox} = 0.10\,\mu\mathrm{m}$, $N_{ox} = 0$, and $V_{FP} = 0$, the 2D procedure selects $(x_c, y_c) \approx (1.148, 0.425)\,\mu\mathrm{m}$ at $r_c = 0.10\,\mu\mathrm{m}$. The centerline-only calculation on the same numerical grid yields approximately 22.301 V, compared with 10.627 V in the 2D corner-excluded search.

Table 2: Strong sensitivity of baseline V_{BD} to chosen artificial corner-exclusion distance.

r_c (nm)	Baseline 2D (V)	x_c (μm)	y_c (μm)
50	8.297	1.193	0.425
100	10.627	1.148	0.425
150	13.313	1.091	0.425

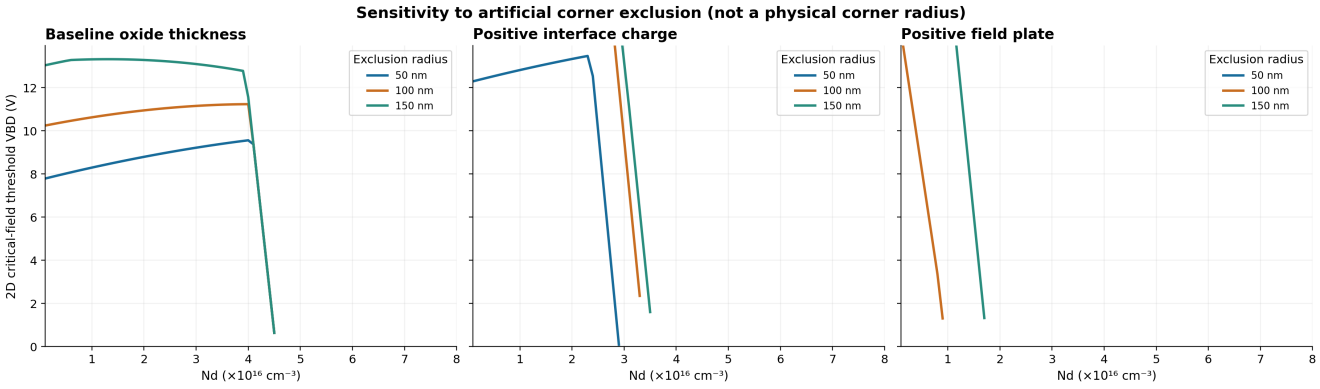


Figure 2: Baseline and selected positive-bias cases under the 50/100/150 nm exclusion conventions. This sensitivity should not be mistaken for a physical trench-radius trend.

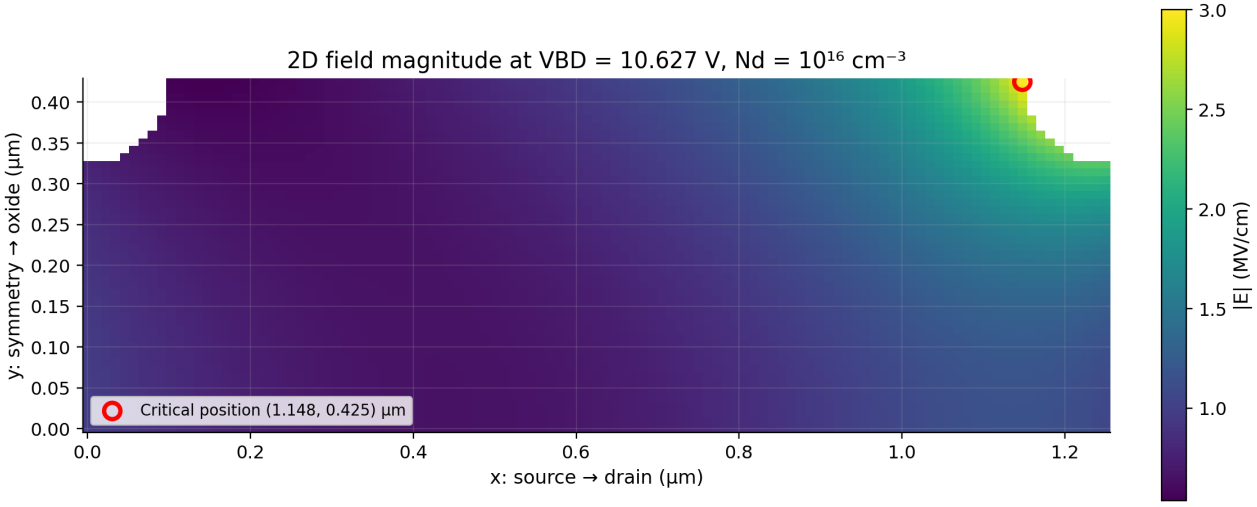


Figure 3: Illustrative field magnitude at the 100 nm corner-excluded baseline threshold. The critical location lies at the oxide interface close to the drain-side exclusion boundary.

8 Numerical reproducibility and limitations

The provided standalone `run_2d_vbd.py` evaluates Eqs. (17)–(24) and exports all plots and numerical CSV tables. To avoid hyperbolic overflow at large n, it evaluates

$$Q_n(y) = \frac{e^{-\alpha_n(a-y)} + e^{-\alpha_n(a+y)}}{[1 + e^{-2\alpha_n a}][1 + b\alpha_n \tanh(\alpha_n a)]} \tag{26}$$

instead of directly constructing two potentially large hyperbolic functions. The derivative $Q_n'(y)$ uses the corresponding difference of exponentials.

Discretization. The location minimization samples a finite grid. Because Fourier differentiation converges more slowly at the oxide interface, the field threshold may oscillate mildly with the number of modes. For the baseline 100 nm case, 420 modes give 10.627 V, while 8000 modes on the same grid give approximately 10.651 V. Additional grid and mode convergence should be checked before relying on any fine numerical detail. The cutoff sensitivity is much larger than these truncation differences.

Physical scope. This linear, uniform-depletion Poisson problem cannot establish measured avalanche breakdown, a realistic trench-corner peak, oxide fringing, trap occupancy dependence, nor the correct actual physical breakdown criterion. E_c here is a fixed illustrative input rather than a measured or doping-calibrated quantity. Any actual predictive device study needs resolved corners, self-consistent depletion and an avalanche-ionization criterion. The present curves are therefore *conditional 2D critical-field thresholds*, not a validated BV_{DSS} prediction.

Truncation cross-check. The field response on the oxide boundary has slowly decaying high-order modes, so the values below are reported as a convergence diagnostic rather than false physical precision:

Sine modes	$V_{\mathrm{BD}}^{(2D,\, r_c=100\,\mathrm{nm})}$ (V)	x_c (μm)
150	10.6337	1.1477
420	10.6267	1.1477
1500	10.6345	1.1477
8000	10.6514	1.1477

This check retains the same 111×47 sampling grid. It does not resolve a physical corner, and it does not imply convergence under shrinking r_c. For model use, both numerical resolution and the evaluation geometry must be specified.

How the three plots should be read. The thickness comparison changes $b = \varepsilon_s t_{\mathrm{ox}}/\varepsilon_{\mathrm{ox}}$ and hence all three response functions in Eq. (17). For charge and field-plate sweeps, t_{ox} is fixed and the changes occur through V_{eff} in Eq. (10). The resulting curves can bend or terminate: the weakest point of the full 2D silicon field can change as the donor concentration changes, and a curve terminates if the zero-drain electric field has already reached E_c.

What is not a fitting parameter. The prescribed E_c is an externally specified threshold; the corner-exclusion radius r_c is an externally specified *evaluation convention*; N_D, V_D, V_{FP}, t_{ox} and N_{ox} are model inputs. No parameter is inferred by fitting any output curve. Nonetheless r_c has a large effect on the 2D maximum, and a realistic rounded trench requires a separate boundary-value solution rather than interpreting the excluded circle as a physical rounded surface.